Ideation Phase

Brainstorm & Idea Prioritization Template

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Team Members: Duddu Vijaya Padma Priya, Kasa Sahitya

Problem Statement Selected: PS-1 — Farmers lack a reliable, data-driven way to choose the most suitable crop based on soil nutrients (Nitrogen, Phosphorus, Potassium) and climate parameters (temperature, humidity, pH, rainfall), leading to poor yield and financial loss.

Step-2: Brainstorm, Idea Listing and Grouping

- Build an ML classification model that takes 7 soil and climate inputs and predicts the best crop to grow
- Use historical agricultural data (N, P, K, temperature, humidity, pH, rainfall, crop label) to train the model
- Create a simple web interface using Streamlit where a farmer can enter values and get instant crop recommendation
- Show top-3 alternate crop suggestions so the farmer has backup options
- Display a confidence score with the prediction to help the farmer trust the recommendation
- Add a simple fertilizer tip based on the predicted crop for extra value

Step-3: Idea Prioritization

Idea	Impact	Feasibility	Priority
ML model (Random Forest) for crop prediction from soil/climate data	High	High	1
Streamlit web app for farmer input and recommendation output	High	High	2
Top-3 alternate crop suggestions	Medium	Medium	3
Fertilizer recommendation tip	Medium	Low	4

Final Selected Approach: Build a Random Forest classification model trained on the Crop Recommendation Dataset, wrapped in a Streamlit web application where users enter N, P, K, temperature, humidity, pH, and rainfall to get an instant crop recommendation with confidence score.